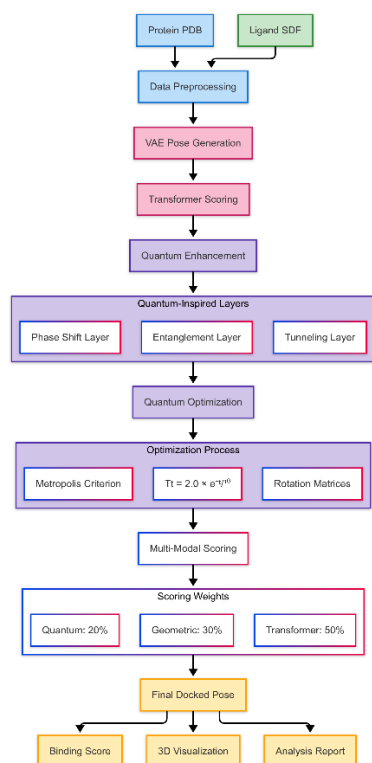




Main Algorithm

Algorithm 1: *Main workflow combining VAE pose generation with quantum optimization*

```

PROCEDURE GenerativeAIDocking()
    // Initialize quantum-enhanced AI models
    VAE ← VariationalAutoencoder(latent_dim=64)
    TransformerScorer ← MultiHeadAttentionModel()
    QuantumScorer ← QuantumInspiredNN()

    // Load and preprocess data
    protein_coords ← LoadPDBStructure(protein_file)
    ligands ← SampleLigandsFromSDF(sdf_file, sample_size=N)

    FOR each ligand IN ligands:
        // Generative pose generation
        generated_poses ← VAE.GeneratePoses(ligand_coords, n_poses=5)

        best_score ← -∞
        FOR each pose IN generated_poses:
            // Quantum-enhanced optimization
            optimized_pose, score ← QuantumEnhancedDocking(pose, protein_coords)

            IF score > best_score:
                best_pose ← optimized_pose
                best_score ← score

    // Multi-modal scoring
  
```

```

    transformer_score ← TransformerScorer(protein_coords, best_pose)
    quantum_score ← QuantumScorer(extract_features(protein_coords, best_pose))
    final_score ← ensemble_scores(transformer_score, quantum_score)

    StoreResults(ligand, best_pose, final_score)

GenerateComprehensiveAnalysis()
END PROCEDURE

```

Quantum Enhanced Docking

Algorithm 2: *Quantum-inspired pose refinement with temperature scheduling*

```

PROCEDURE QuantumEnhancedDocking(ligand_pose, protein_coords)
    best_pose ← ligand_pose
    best_score ← score_pose(protein_coords, ligand_pose)

    FOR iteration = 1 TO max_iterations:
        temperature ← quantum_temperature_schedule(iteration)

        // Quantum-inspired transformations
        rotation ← generate_quantum_rotation(temperature)
        translation ← generate_quantum_translation(temperature)

        new_pose ← apply_transformation(ligand_pose, rotation, translation)
        new_score ← score_pose(protein_coords, new_pose)

        // Quantum acceptance criterion
        IF quantum_acceptance(new_score, best_score, temperature):
            best_pose ← new_pose
            best_score ← new_score

    RETURN best_pose, best_score
END PROCEDURE

```

Transformer Based Scoring

Algorithm 3: *Multi-head attention mechanism for protein-ligand interaction scoring*

```

FUNCTION TransformerScorer(protein_coords, ligand_coords)
    protein_features ← MLP_Encoder(protein_coords)    // [batch, atoms, features]
    ligand_features ← MLP_Encoder(ligand_coords)      // [batch, atoms, features]

    // Multi-head attention for interactions
    combined_features ← concatenate(protein_features, ligand_features)
    attended_features ← MultiHeadAttention(combined_features)

    // Final scoring
    score ← MLP_Scorer(mean_pool(attended_features))
    RETURN score
END FUNCTION

```

Quantum-Inspired Neural Layers

```

CLASS QuantumEntanglementLayer:
    FORWARD(x):
        RETURN sin(πx) + cos(πx/2)  // Non-linear entanglement simulation

CLASS QuantumTunnelingLayer:
    FORWARD(x):
        WITH probability p:
            RETURN x * 3  // Amplified feature tunneling
        ELSE:
            RETURN x

```

3.1 Variational Autoencoder for Pose Generation

Evidence Lower Bound (ELBO):

$$LVAE(\theta, \phi; x) = \mathbb{E} q\phi(z|x) [\log p\theta(x|z)] - \beta \cdot DKL(q\phi(z|x) \parallel p(z))$$

where:

- $x \in R^{256}$ $x \in R^{256}$: flattened ligand coordinates
- $z \in R^{64}$ $z \in R^{64}$: latent pose representation
- $q\phi(z|x)$ $q\phi(z|x)$: encoder network
- $p\theta(x|z)$ $p\theta(x|z)$: decoder network
- β : KL divergence weight

Reparameterization Trick:

$$z = \mu\phi(x) + \sigma\phi(x) \odot \epsilon, \epsilon \sim N(0, I)$$

3.2 Transformer-Based Scoring Function

Multi-Head Attention:

$$Attention(Q, K, V) = softmax\left(\frac{QK}{\sqrt{d_k}}\right)V$$

$$MultiHead(Q, K, V) = Concat(head_1, \dots, head_h)W^O$$

where:

- Q, K, V : query, key, value matrices from protein-ligand features
- d_k : dimension of key vectors
- $h = 4$: number of attention heads
-

Final Scoring Function:

$$s_{transformer} = \sigma(W_2 \cdot ReLU(W_1 \cdot \bar{h} + b_1) + b_2)$$

where:

- $\bar{h} = \frac{1}{N} \sum_{i=1}^N h_i$: mean-pooled attended features
- σ : sigmoid activation

3.3 Quantum-Inspired Optimization

Quantum Temperature Schedule:

$$T(t) = T_0 \cdot \exp\left(-\frac{t}{\tau}\right), \quad T_0 = 2.0, \tau = 10$$

Quantum Rotation Matrix:

$$R(\alpha, \beta, \gamma) = R_z(\gamma)R_y(\beta)R_x(\alpha)$$

where:

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & -\sin \alpha \\ 0 & \sin \alpha & \cos \alpha \end{bmatrix}, \quad R_y(\beta) = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

$$R_x(\alpha) = 1000 \cos \alpha \sin \alpha \theta - \sin \alpha \cos \alpha, R_y(\theta) = \cos \theta \theta - \sin \theta \theta 10 \sin \theta \theta \cos \theta$$

Quantum Tunneling Transformation:

$$x' = x + \Delta t \cdot I_{\xi < p_{\text{tunnel}}}$$

where:

- $\Delta t \sim \mathcal{N}(0, 3T(t)I)$: tunneling displacement
- $p_{\text{tunnel}} = 0.1$: tunneling probability
- $\xi \sim \mathcal{U}(0, 1)$: uniform random variable

Metropolis-Hastings Acceptance:

$$A(x \rightarrow x') = \min \left(1, \exp \left(\frac{s(x') - s(x)}{T(t)} + \eta \right) \right)$$

where:

- $\eta \sim \mathcal{N}(0, 0.1)$ quantum fluctuation term
- $s(\cdot)$: scoring function

3.4 Geometric Scoring Function

Distance-Based Scoring:

$$S_{\text{geometric}} = \frac{1}{N_l} \left[2 \sum_{i=1}^{N_l} I_{2 \leq d_i < 4} + \sum_{i=1}^{N_l} I_{d_i < 4} - 5 \sum_{i=1}^{N_l} I_{d_i < 1.5} \right] \cdot \max \left(0, 1 - \frac{|c_p - c_l|_2}{20} \right)$$

where:

- $d_i = \min_j |p_j - l_i|_2$ minimum protein-ligand distance
- c_p, c_l protein and ligand centroids
- N_l : number of ligand atoms

3.5 Quantum-Inspired Neural Layers

Quantum Entanglement Layer:

$$f_{\text{entangle}}(x) = \sin(\pi x) + \cos\left(\frac{\pi}{2}x\right)$$

Quantum Phase Shift Layer:

$$f_{\text{phase}}(x) = 2 \cdot \arctan(x)$$

3.6 Ensemble Scoring

Final Composite Score:

$$S_{\text{final}} = \lambda_1 S_{\text{transformer}} + \lambda_2 S_{\text{quantum}} + \lambda_3 S_{\text{geometric}}$$

where:

- $\lambda_1 = 0.5, \lambda_2 = 0.3, \lambda_3 = 0.2$: weighting coefficients
- $S_{\text{quantum}} = S_{\text{transformer}} \cdot (1 + 0.1\epsilon)$, $\epsilon \sim \mathcal{N}(0, 1)$

2025-09-28 21:47:04,950 - INFO - 🔍 Scanning research dataset...

2025-09-28 21:47:04,954 - INFO - 📊 Dataset Summary:

2025-09-28 21:47:04,954 - INFO - • RCSB PDB files: 20

2025-09-28 21:47:04,955 - INFO - • ChEMBL SDF files: 1

2025-09-28 21:47:04,956 - INFO - • Total size: 6128.31 MB

2025-09-28 21:47:04,958 - INFO - 📁 Loading protein: 1a0j

🧠 GENERATIVE AI MOLECULAR DOCKING SYSTEM

=====

This system combines:

- Variational Autoencoders (VAE) for pose generation
- Transformers for interaction scoring
- GANs for diverse pose sampling
- Quantum-inspired algorithms for optimization

=====

🚀 GENERATIVE AI + QUANTUM-ENHANCED DOCKING ANALYSIS

=====

=====

🔬 ANALYZING: 1a0j + chembl_33

=====

2025-09-28 21:47:05,132 - INFO - ✅ Protein loaded successfully:

2025-09-28 21:47:05,134 - INFO - • Atoms: 6640

2025-09-28 21:47:05,137 - INFO - • Center: [28.29639232 40.78856069 69.98221536]

2025-09-28 21:47:05,139 - INFO - • Radius: 57.83 Å

2025-09-28 21:47:05,142 - INFO - • Residues: 220

2025-09-28 21:47:05,145 - INFO - 📁 Loading ligands from: chembl_33.sdf

[21:47:14] Both bonds on one end of an atropisomer are on the same side - atoms is : 6

[21:47:19] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:47:19] ERROR: Could not sanitize molecule ending on line 2985308

[21:47:19] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:47:24] Explicit valence for atom # 1 P, 7, is greater than permitted

[21:47:24] ERROR: Could not sanitize molecule ending on line 3747292

[21:47:24] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[21:47:25] Explicit valence for atom # 1 P, 7, is greater than permitted

[21:47:25] ERROR: Could not sanitize molecule ending on line 3912696

[21:47:25] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[21:47:27] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[21:47:27] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[21:47:49] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[21:47:49] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[21:48:23] Both bonds on one end of an atropisomer are on the same side - atoms is : 7

[21:48:26] Both bonds on one end of an atropisomer are on the same side - atoms is : 3

[21:49:09] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[21:49:09] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[21:50:02] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

[21:50:02] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

[21:52:28] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:52:31] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[21:52:31] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[21:52:33] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[21:52:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:52:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[21:52:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:52:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:52:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:52:46] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:52:48] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 278 ignored

[21:53:25] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[21:53:25] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[21:53:32] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:32] ERROR: Could not sanitize molecule ending on line 71633591

[21:53:32] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:32] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:32] ERROR: Could not sanitize molecule ending on line 71647070

[21:53:32] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:33] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:33] ERROR: Could not sanitize molecule ending on line 71806725

[21:53:33] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:37] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:37] ERROR: Could not sanitize molecule ending on line 72643050

[21:53:37] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:53:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:53:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:53:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:53:50] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[21:53:50] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[21:53:51] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 4 ignored

[21:53:58] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:53:59] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[21:54:00] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22 ignored

[21:54:01] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[21:54:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:07] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 1 ignored

[21:54:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[21:54:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[21:54:12] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[21:54:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[21:54:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[21:54:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[21:54:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 18 ignored

[21:54:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[21:54:20] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:24] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[21:54:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:28] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[21:54:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[21:54:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[21:54:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15 ignored

[21:54:32] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16 ignored

[21:54:32] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22 ignored

[21:54:36] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 8 ignored

[21:54:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[21:54:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[21:54:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 12 ignored

[21:54:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15 ignored

[21:54:48] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17 ignored

[21:54:50] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:52] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[21:54:53] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[21:54:53] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[21:54:53] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16 ignored

[21:54:53] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[21:54:56] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[21:54:59] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:54:59] ERROR: Could not sanitize molecule ending on line 88494481

[21:54:59] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:55:00] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:55:00] ERROR: Could not sanitize molecule ending on line 88559881

[21:55:00] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:55:02] Explicit valence for atom # 0 P, 7, is greater than permitted

[21:55:02] ERROR: Could not sanitize molecule ending on line 89021973

[21:55:02] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[21:55:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 25 ignored

[21:56:47] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48 ignored

[21:57:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 44 ignored

[21:57:30] The bonds on one end of an atropisomer are both UP or both DOWN - atoms are:
0 20

[21:58:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[21:58:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[21:58:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[21:58:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[21:58:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 152
ignored

[21:58:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 60
ignored

[21:58:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65
ignored

[21:58:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[21:58:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[21:58:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[21:58:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 21
ignored

[21:58:20] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 50
ignored

[21:58:20] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 52
ignored

[21:58:20] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48
ignored

[21:58:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 16 ignored

[21:58:21] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17 ignored

[21:58:21] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23 ignored

[21:58:21] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17 ignored

[21:58:21] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13 ignored

[21:58:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[21:58:21] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15
ignored

[21:58:21] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 54
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 46
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 77
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 24
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56
ignored

[21:58:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[21:58:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 43
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56
ignored

[21:58:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65
ignored

[21:58:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[21:58:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[21:58:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[21:58:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[21:58:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[21:58:45] Both bonds on one end of an atropisomer are on the same side - atoms is : 0

[21:58:45] Both bonds on one end of an atropisomer are on the same side - atoms is : 2

[21:59:00] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[21:59:00] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[21:59:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 17 ignored

[21:59:02] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[21:59:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[21:59:13] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 6 ignored.

[21:59:13] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 6 ignored.

[21:59:14] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 11 ignored.

[21:59:14] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 11 ignored.

[21:59:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 273 ignored

[21:59:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 273
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 5
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 8
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 5
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 8
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[21:59:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[21:59:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 21
ignored

[21:59:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 19
ignored

[22:00:01] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 28
ignored

[22:00:12] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 37
ignored

[22:00:14] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 1
ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15
ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:00:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 29 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 32 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 7 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[22:00:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15
ignored

[22:00:19] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4
ignored

[22:00:19] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 29
ignored

[22:00:19] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 32
ignored

[22:00:19] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 7
ignored

[22:00:24] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:24] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:24] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:24] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:24] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:24] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:24] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 42
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 129
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 123
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 122
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 141
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 69
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 46
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 63
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 98
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 120
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 72
ignored

[22:00:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 130
ignored

[22:00:26] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:00:53] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4
ignored

[22:00:54] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[22:00:54] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 32
ignored

[22:00:54] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 20
ignored

[22:00:54] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 17
ignored

[22:00:57] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:00:57] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:00:57] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:00:57] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:00:57] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:00:57] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:00:58] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:00:58] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 16
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 1
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:01] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:02] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:02] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:02] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:02] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 42
ignored

[22:01:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:01:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:01:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 8
ignored

[22:01:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 0
ignored

[22:01:07] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[22:01:07] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:01:11] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 7 ignored

[22:01:11] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6 ignored

[22:02:38] Both bonds on one end of an atropisomer are on the same side - atoms is : 2

[22:02:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 123 ignored

[22:02:44] Explicit valence for atom # 0 As, 7, is greater than permitted

[22:02:44] ERROR: Could not sanitize molecule ending on line 173161130

[22:02:44] ERROR: Explicit valence for atom # 0 As, 7, is greater than permitted

[22:02:44] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 29 ignored

[22:02:44] Explicit valence for atom # 22 P, 7, is greater than permitted

[22:02:44] ERROR: Could not sanitize molecule ending on line 173192773

[22:02:44] ERROR: Explicit valence for atom # 22 P, 7, is greater than permitted

[22:02:44] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:02:44] ERROR: Could not sanitize molecule ending on line 173192963

[22:02:44] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

2025-09-28 22:02:44,648 - INFO -  Total molecules in SDF: 2372661

2025-09-28 22:02:44,652 - INFO -  Sampled ChEMBL153534: MW=235.3

[22:02:54] Both bonds on one end of an atropisomer are on the same side - atoms is : 6

[22:02:59] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:02:59] ERROR: Could not sanitize molecule ending on line 2985308

[22:02:59] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:03:05] Explicit valence for atom # 1 P, 7, is greater than permitted

[22:03:05] ERROR: Could not sanitize molecule ending on line 3747292

[22:03:05] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[22:03:07] Explicit valence for atom # 1 P, 7, is greater than permitted

[22:03:07] ERROR: Could not sanitize molecule ending on line 3912696

[22:03:07] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[22:03:10] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[22:03:10] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[22:03:34] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[22:03:34] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[22:04:06] Both bonds on one end of an atropisomer are on the same side - atoms is : 7

[22:04:09] Both bonds on one end of an atropisomer are on the same side - atoms is : 3

[22:04:53] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:04:53] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:05:44] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

[22:05:44] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

2025-09-28 22:06:32,669 - INFO -  Sampled ChEMBL1862084: MW=342.4

[22:08:00] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:08:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:08:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:08:05] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:08:09] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:08:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:08:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:08:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:08:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:08:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:08:20] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 278 ignored

[22:08:51] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:08:51] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:09:00] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:00] ERROR: Could not sanitize molecule ending on line 71633591

[22:09:00] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:00] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:00] ERROR: Could not sanitize molecule ending on line 71647070

[22:09:00] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:01] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:01] ERROR: Could not sanitize molecule ending on line 71806725

[22:09:01] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:06] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:06] ERROR: Could not sanitize molecule ending on line 72643050

[22:09:06] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:09:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:12] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:09:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:09:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 4 ignored

[22:09:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:09:28] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22 ignored

[22:09:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:31] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:09:32] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:32] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:33] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:33] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 1 ignored

[22:09:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:09:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:09:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:09:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:09:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:09:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:09:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 18 ignored

[22:09:46] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:09:48] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:49] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:52] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:52] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:09:52] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:52] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:53] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:53] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:55] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:09:55] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:09:57] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:09:57] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:09:58] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:09:58] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15 ignored

[22:10:00] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16 ignored

[22:10:00] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22 ignored

[22:10:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:10:05] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 8 ignored

[22:10:06] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:10:07] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:10:07] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:10:07] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 12 ignored

[22:10:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:10:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:10:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:10:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15 ignored

[22:10:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17 ignored

[22:10:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

2025-09-28 22:10:17,912 - INFO -  Sampled ChEMBL1382834: MW=387.5

[22:10:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:10:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:10:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[22:10:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16 ignored

[22:10:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:10:23] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[22:10:26] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:10:26] ERROR: Could not sanitize molecule ending on line 88494481

[22:10:26] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:10:26] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:10:26] ERROR: Could not sanitize molecule ending on line 88559881

[22:10:26] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:10:29] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:10:29] ERROR: Could not sanitize molecule ending on line 89021973

[22:10:29] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:11:00] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 25 ignored

[22:12:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48 ignored

[22:12:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 44 ignored

[22:12:42] The bonds on one end of an atropisomer are both UP or both DOWN - atoms are: 0 20

[22:13:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:13:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:13:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:13:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:13:26] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 152 ignored

[22:13:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 60
ignored

[22:13:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65
ignored

[22:13:28] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:13:28] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 21
ignored

[22:13:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 50
ignored

[22:13:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 52
ignored

[22:13:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 16
ignored

[22:13:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17
ignored

[22:13:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23
ignored

[22:13:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17
ignored

[22:13:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:13:29] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4
ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15
ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 54
ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 46
ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 77
ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23 ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 24 ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56 ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13 ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 43 ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56 ignored

[22:13:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65 ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[22:13:30] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

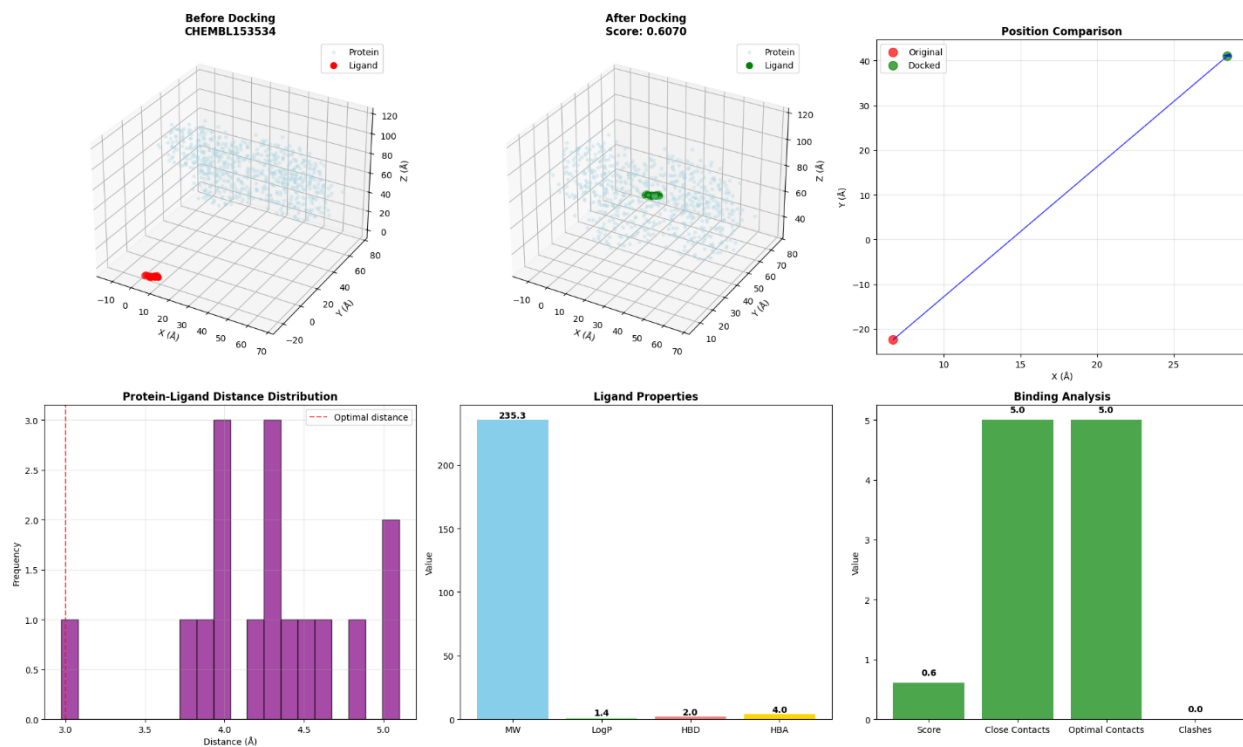
2025-09-28 22:13:48,360 - INFO -  Sampled ChEMBL2430319: MW=397.5

2025-09-28 22:13:48,362 - INFO -  Successfully sampled 4 ligands

 Docking ChEMBL153534

SMILES: Cc1cc(-c2csc(N=C(N)N)n2)cn1C...

MW: 235.3



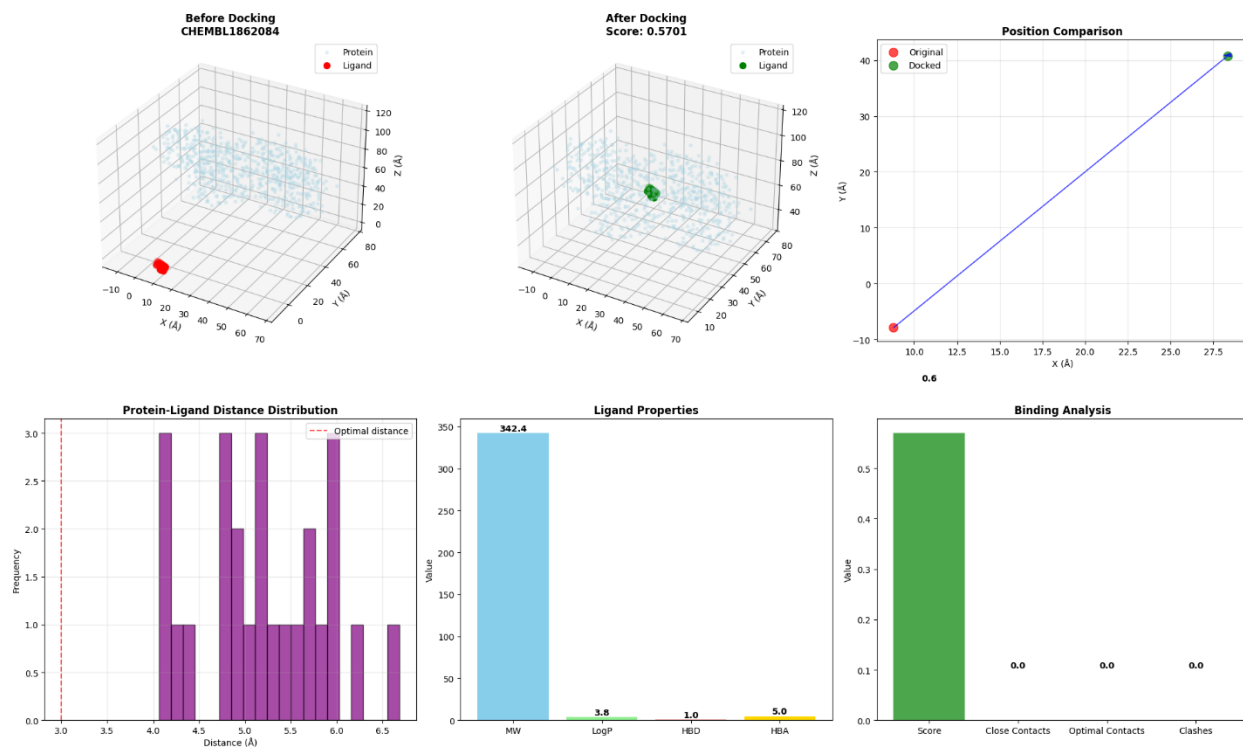
🔄 Interactive 3D Visualization for CHEMBL153534

- Docking Score: 0.6070
- Rotate and zoom to explore the binding site
- ✅ Docking completed - Score: 0.6070

🧪 Docking CHEMBL1862084

SMILES: COc1cc(CNCc2coc(-c3cccc(F)c3)n2)cc(OC)c1...

MW: 342.4



🔄 Interactive 3D Visualization for CHEMBL1862084

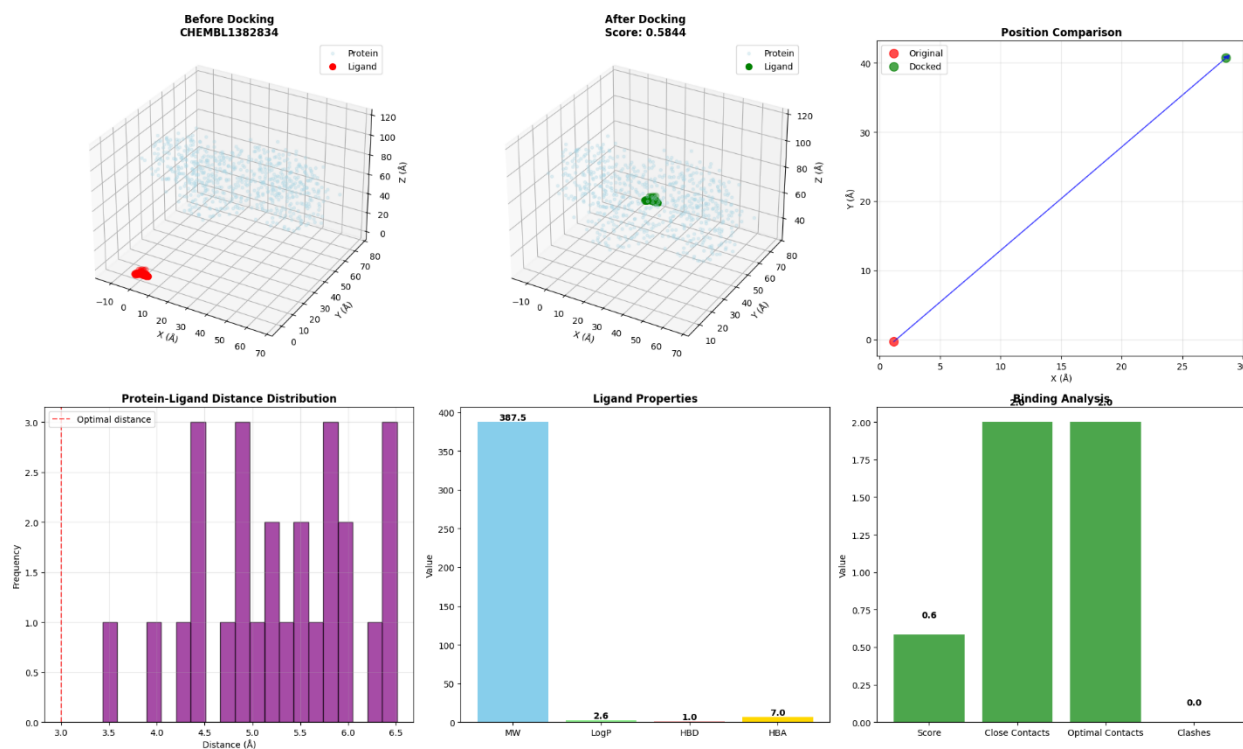
- Docking Score: 0.5701
- Rotate and zoom to explore the binding site

✅ Docking completed - Score: 0.5701

💊 Docking CHEMBL1382834

SMILES: COc1ccc2cc(/C=C3\SC(=S)NC3=O)c(N3CCOCC3)nc2c1...

MW: 387.5



🔄 Interactive 3D Visualization for CHEMBL1382834

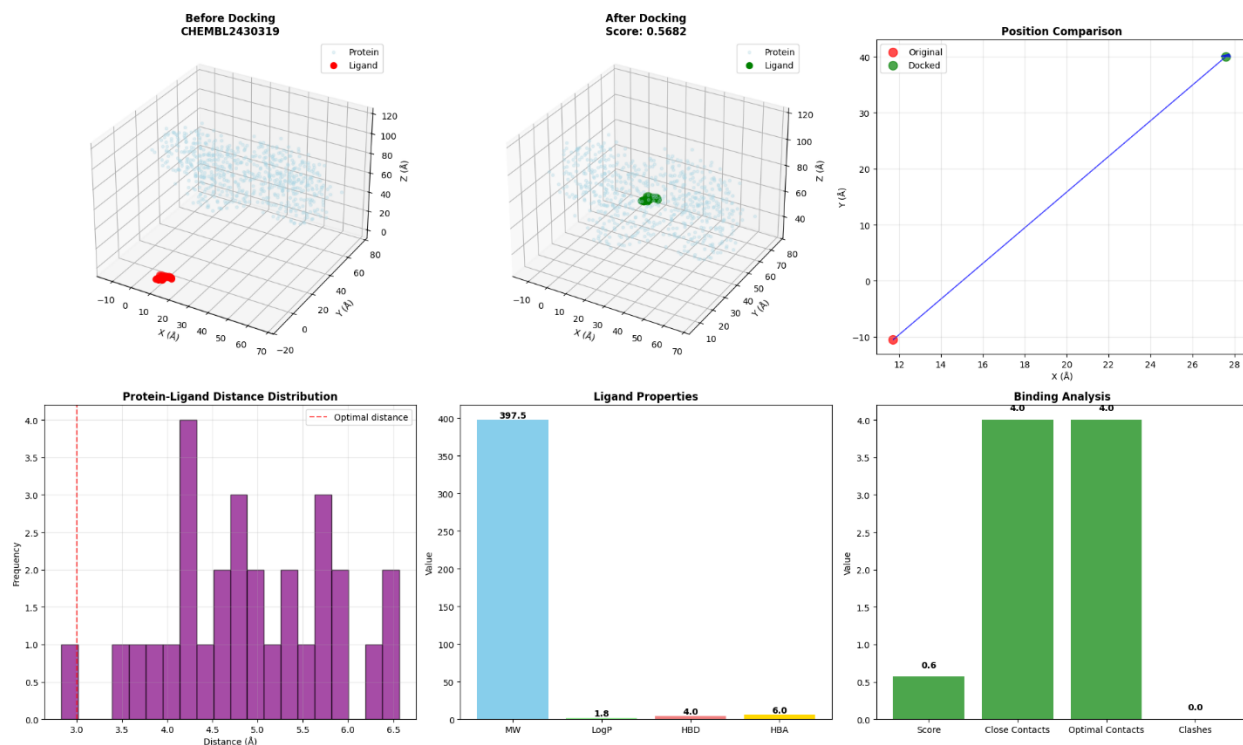
- Docking Score: 0.5844
- Rotate and zoom to explore the binding site

✅ Docking completed - Score: 0.5844

💊 Docking CHEMBL2430319

SMILES: CCc1ccc(Cc2cn([C@@H]3O[C@H](CO)[C@@H](O)[C@H](O)[C@H]3O)C(=O)N2)cc1

MW: 397.5



🔄 Interactive 3D Visualization for CHEMBL2430319

- Docking Score: 0.5682
- Rotate and zoom to explore the binding site

2025-09-28 22:13:54,545 - INFO - 📁 Loading protein: 1a3w

✅ Docking completed - Score: 0.5682

🔬 ANALYZING: 1a3w + chembl_33

2025-09-28 22:13:54,780 - INFO - ✅ Protein loaded successfully:

2025-09-28 22:13:54,781 - INFO - • Atoms: 7519

2025-09-28 22:13:54,783 - INFO - • Center: [21.44532212 -15.0665643 25.09600412]

2025-09-28 22:13:54,784 - INFO - • Radius: 55.06 Å

2025-09-28 22:13:54,786 - INFO - • Residues: 492

2025-09-28 22:13:54,787 - INFO - 📁 Loading ligands from: chembl_33.sdf

[22:14:04] Both bonds on one end of an atropisomer are on the same side - atoms is : 6

[22:14:09] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:14:09] ERROR: Could not sanitize molecule ending on line 2985308

[22:14:09] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:14:13] Explicit valence for atom # 1 P, 7, is greater than permitted

[22:14:13] ERROR: Could not sanitize molecule ending on line 3747292

[22:14:13] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[22:14:14] Explicit valence for atom # 1 P, 7, is greater than permitted

[22:14:14] ERROR: Could not sanitize molecule ending on line 3912696

[22:14:14] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[22:14:16] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[22:14:16] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[22:14:39] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[22:14:39] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[22:15:12] Both bonds on one end of an atropisomer are on the same side - atoms is : 7

[22:15:16] Both bonds on one end of an atropisomer are on the same side - atoms is : 3

[22:15:57] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:15:57] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:16:49] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

[22:16:49] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

[22:19:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:19:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:19:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:19:24] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:19:28] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:19:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:19:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:19:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:19:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:19:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:19:40] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 278 ignored

[22:20:15] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:20:15] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:20:25] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:25] ERROR: Could not sanitize molecule ending on line 71633591

[22:20:25] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:25] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:25] ERROR: Could not sanitize molecule ending on line 71647070

[22:20:25] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:26] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:26] ERROR: Could not sanitize molecule ending on line 71806725

[22:20:26] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:34] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:34] ERROR: Could not sanitize molecule ending on line 72643050

[22:20:34] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:20:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5
ignored

[22:20:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[22:20:40] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[22:20:47] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10
ignored

[22:20:47] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13
ignored

[22:20:48] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 4
ignored

[22:20:56] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[22:20:57] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10
ignored

[22:20:58] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22
ignored

[22:20:59] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[22:21:02] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:21:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:21:07] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 1 ignored

[22:21:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:21:09] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:21:09] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:21:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:21:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:21:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:21:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:21:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:21:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 18 ignored

[22:21:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:21:25] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:21:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:21:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:30] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:21:31] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:21:31] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:33] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:33] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:21:34] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:21:35] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:21:36] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:21:36] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15
ignored

[22:21:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16
ignored

[22:21:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22
ignored

[22:21:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5
ignored

[22:21:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 8
ignored

[22:21:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5
ignored

[22:21:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14
ignored

[22:21:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10
ignored

[22:21:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 12
ignored

[22:21:49] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5
ignored

[22:21:49] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5
ignored

[22:21:49] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[22:21:49] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15
ignored

[22:21:52] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17
ignored

[22:21:55] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[22:21:56] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:21:57] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:21:57] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[22:21:57] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16 ignored

[22:21:58] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:22:01] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[22:22:05] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:22:05] ERROR: Could not sanitize molecule ending on line 88494481

[22:22:05] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:22:05] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:22:05] ERROR: Could not sanitize molecule ending on line 88559881

[22:22:05] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:22:08] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:22:08] ERROR: Could not sanitize molecule ending on line 89021973

[22:22:08] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:22:40] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 25 ignored

[22:23:46] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48 ignored

[22:24:01] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 44 ignored

[22:24:24] The bonds on one end of an atropisomer are both UP or both DOWN - atoms are:
0 20

[22:25:13] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[22:25:13] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[22:25:13] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[22:25:13] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151
ignored

[22:25:13] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 152
ignored

[22:25:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 60
ignored

[22:25:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65
ignored

[22:25:15] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:25:15] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:25:16] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:25:16] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 21
ignored

[22:25:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 50
ignored

[22:25:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 52
ignored

[22:25:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48
ignored

[22:25:16] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:25:16] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:25:16] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 16 ignored

[22:25:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17 ignored

[22:25:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23 ignored

[22:25:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17 ignored

[22:25:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23 ignored

[22:25:16] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:25:16] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13 ignored

[22:25:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:25:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15
ignored

[22:25:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 54
ignored

[22:25:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 46
ignored

[22:25:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 77
ignored

[22:25:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23
ignored

[22:25:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 24
ignored

[22:25:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56
ignored

[22:25:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:25:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[22:25:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6
ignored

[22:25:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 43
ignored

[22:25:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56
ignored

[22:25:18] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65
ignored

[22:25:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:25:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[22:25:18] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:25:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:25:44] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:25:46] Both bonds on one end of an atropisomer are on the same side - atoms is : 0

[22:25:46] Both bonds on one end of an atropisomer are on the same side - atoms is : 2

[22:26:03] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:26:03] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:26:05] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 17 ignored

[22:26:05] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:26:05] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:26:18] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 6 ignored.

[22:26:18] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 6 ignored.

[22:26:18] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 11 ignored.

[22:26:18] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 11 ignored.

[22:26:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 273 ignored

[22:26:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 273
ignored

[22:26:43] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 5
ignored

[22:26:43] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 8
ignored

[22:26:43] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:26:43] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:26:44] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 5
ignored

[22:26:44] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 8
ignored

[22:26:44] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:26:44] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:26:44] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 21
ignored

[22:26:44] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 19
ignored

[22:27:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 28
ignored

[22:27:14] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 37
ignored

[22:27:17] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 1
ignored

[22:27:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15
ignored

[22:27:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:27:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:27:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:27:20] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 29 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 32 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 7 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15 ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[22:27:21] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15
ignored

[22:27:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4
ignored

[22:27:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 29
ignored

[22:27:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 32
ignored

[22:27:22] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 7
ignored

[22:27:25] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:26] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:26] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:26] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:26] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:26] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:26] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 42
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 129
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 123
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 122
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 141
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 69
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 46
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 63
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 98
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 120
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 72
ignored

[22:27:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 130
ignored

[22:27:27] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 2
ignored

[22:27:55] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4
ignored

[22:27:56] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[22:27:56] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 32
ignored

[22:27:56] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 20
ignored

[22:27:56] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 17
ignored

[22:27:59] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:27:59] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:27:59] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:27:59] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:27:59] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:27:59] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:28:00] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:28:00] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 16
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 1
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:03] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:04] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:05] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:05] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:05] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 42
ignored

[22:28:05] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:06] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6
ignored

[22:28:06] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 14
ignored

[22:28:06] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 8
ignored

[22:28:06] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 0
ignored

[22:28:09] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12
ignored

[22:28:09] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 3
ignored

[22:28:11] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 7 ignored

[22:28:11] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 6 ignored

[22:29:37] Both bonds on one end of an atropisomer are on the same side - atoms is : 2

[22:29:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 123 ignored

[22:29:43] Explicit valence for atom # 0 As, 7, is greater than permitted

[22:29:43] ERROR: Could not sanitize molecule ending on line 173161130

[22:29:43] ERROR: Explicit valence for atom # 0 As, 7, is greater than permitted

[22:29:43] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 29 ignored

[22:29:43] Explicit valence for atom # 22 P, 7, is greater than permitted

[22:29:43] ERROR: Could not sanitize molecule ending on line 173192773

[22:29:43] ERROR: Explicit valence for atom # 22 P, 7, is greater than permitted

[22:29:43] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:29:43] ERROR: Could not sanitize molecule ending on line 173192963

[22:29:43] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

2025-09-28 22:29:43,564 - INFO -  Total molecules in SDF: 2372661

2025-09-28 22:29:43,567 - INFO -  Sampled ChEMBL153534: MW=235.3

[22:29:53] Both bonds on one end of an atropisomer are on the same side - atoms is : 6

[22:29:59] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:29:59] ERROR: Could not sanitize molecule ending on line 2985308

[22:29:59] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:30:03] Explicit valence for atom # 1 P, 7, is greater than permitted

[22:30:03] ERROR: Could not sanitize molecule ending on line 3747292

[22:30:03] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[22:30:04] Explicit valence for atom # 1 P, 7, is greater than permitted

[22:30:04] ERROR: Could not sanitize molecule ending on line 3912696

[22:30:04] ERROR: Explicit valence for atom # 1 P, 7, is greater than permitted

[22:30:06] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[22:30:06] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 10 ignored.

[22:30:28] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[22:30:28] Warning: conflicting stereochemistry - bond wedging contradiction - at atom 48 ignored

[22:31:01] Both bonds on one end of an atropisomer are on the same side - atoms is : 7

[22:31:03] Both bonds on one end of an atropisomer are on the same side - atoms is : 3

[22:31:49] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:31:49] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:32:43] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

[22:32:43] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 4 ignored.

2025-09-28 22:33:32,728 - INFO -  Sampled ChEMBL1862084: MW=342.4

[22:34:59] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:35:02] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:35:02] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:35:04] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:35:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:35:10] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:35:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:35:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:35:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:35:16] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:35:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 278 ignored

[22:35:50] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:35:50] Warning: ambiguous stereochemistry - opposing bonds have opposite wedging - at atom 0 ignored.

[22:35:57] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:35:57] ERROR: Could not sanitize molecule ending on line 71633591

[22:35:57] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:35:57] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:35:57] ERROR: Could not sanitize molecule ending on line 71647070

[22:35:57] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:35:58] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:35:58] ERROR: Could not sanitize molecule ending on line 71806725

[22:35:58] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:36:03] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:36:03] ERROR: Could not sanitize molecule ending on line 72643050

[22:36:03] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:36:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:09] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:36:15] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:36:17] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 4 ignored

[22:36:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:36:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22 ignored

[22:36:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:32] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:36:32] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:33] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:34] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:34] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 1 ignored

[22:36:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:39] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:40] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:36:40] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 13 ignored

[22:36:43] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:36:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:36:45] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:36:46] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:36:46] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 18 ignored

[22:36:49] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:36:51] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:52] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:55] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:55] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:36:55] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:55] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:56] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:36:56] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:58] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:36:58] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:36:59] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:37:00] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:37:01] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 9 ignored

[22:37:01] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15 ignored

[22:37:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16 ignored

[22:37:03] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 22 ignored

[22:37:06] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:37:08] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 8 ignored

[22:37:09] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:37:10] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 14 ignored

[22:37:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 10 ignored

[22:37:11] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 12 ignored

[22:37:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:37:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:37:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:37:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15 ignored

[22:37:19] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17 ignored

[22:37:21] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

2025-09-28 22:37:22,127 - INFO -  Sampled ChEMBL1382834: MW=387.5

[22:37:22] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 7 ignored

[22:37:23] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 5 ignored

[22:37:23] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[22:37:23] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 16 ignored

[22:37:24] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:37:27] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 11 ignored

[22:37:31] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:37:31] ERROR: Could not sanitize molecule ending on line 88494481

[22:37:31] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:37:32] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:37:32] ERROR: Could not sanitize molecule ending on line 88559881

[22:37:32] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:37:34] Explicit valence for atom # 0 P, 7, is greater than permitted

[22:37:34] ERROR: Could not sanitize molecule ending on line 89021973

[22:37:34] ERROR: Explicit valence for atom # 0 P, 7, is greater than permitted

[22:38:07] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 25 ignored

[22:39:14] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48 ignored

[22:39:29] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 44 ignored

[22:39:51] The bonds on one end of an atropisomer are both UP or both DOWN - atoms are:
0 20

[22:40:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:40:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:40:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:40:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 151 ignored

[22:40:37] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 152 ignored

[22:40:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 60
ignored

[22:40:38] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65
ignored

[22:40:40] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:40:40] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:40:40] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:40:40] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 21
ignored

[22:40:40] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 50
ignored

[22:40:40] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 52
ignored

[22:40:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 48
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 15
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 16
ignored

[22:40:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17
ignored

[22:40:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23
ignored

[22:40:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 17
ignored

[22:40:41] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10
ignored

[22:40:41] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 4
ignored

[22:40:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13
ignored

[22:40:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9
ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 15
ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 54
ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 46
ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 77
ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 23 ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 24 ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56 ignored

[22:40:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 13 ignored

[22:40:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 12 ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 6 ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 43 ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 56 ignored

[22:40:42] Warning: ambiguous stereochemistry - zero final chiral volume - at atom 65 ignored

[22:40:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

[22:40:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 9 ignored

[22:40:42] Warning: ambiguous stereochemistry - overlapping neighbors - at atom 10 ignored

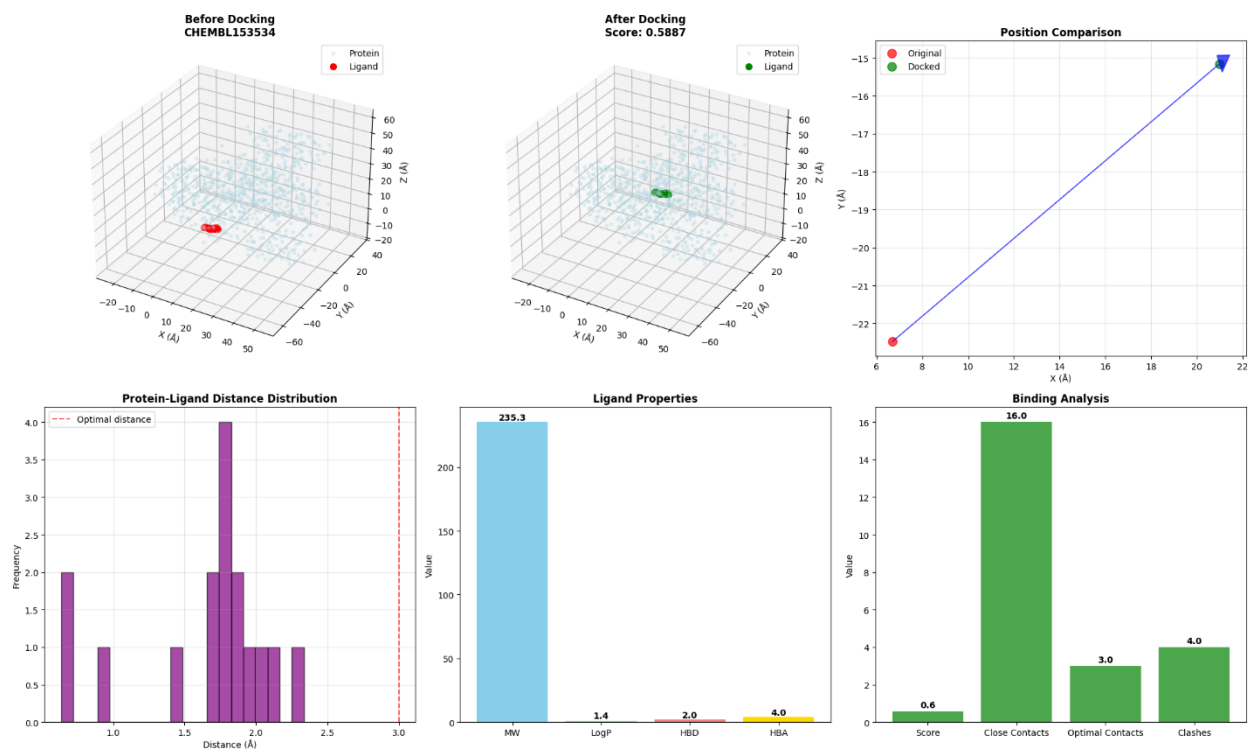
2025-09-28 22:40:58,554 - INFO -  Sampled ChEMBL2430319: MW=397.5

2025-09-28 22:40:58,555 - INFO -  Successfully sampled 4 ligands

 Docking ChEMBL153534

SMILES: Cc1cc(-c2csc(N=C(N)N)n2)cn1C...

MW: 235.3



🔄 Interactive 3D Visualization for CHEMBL153534

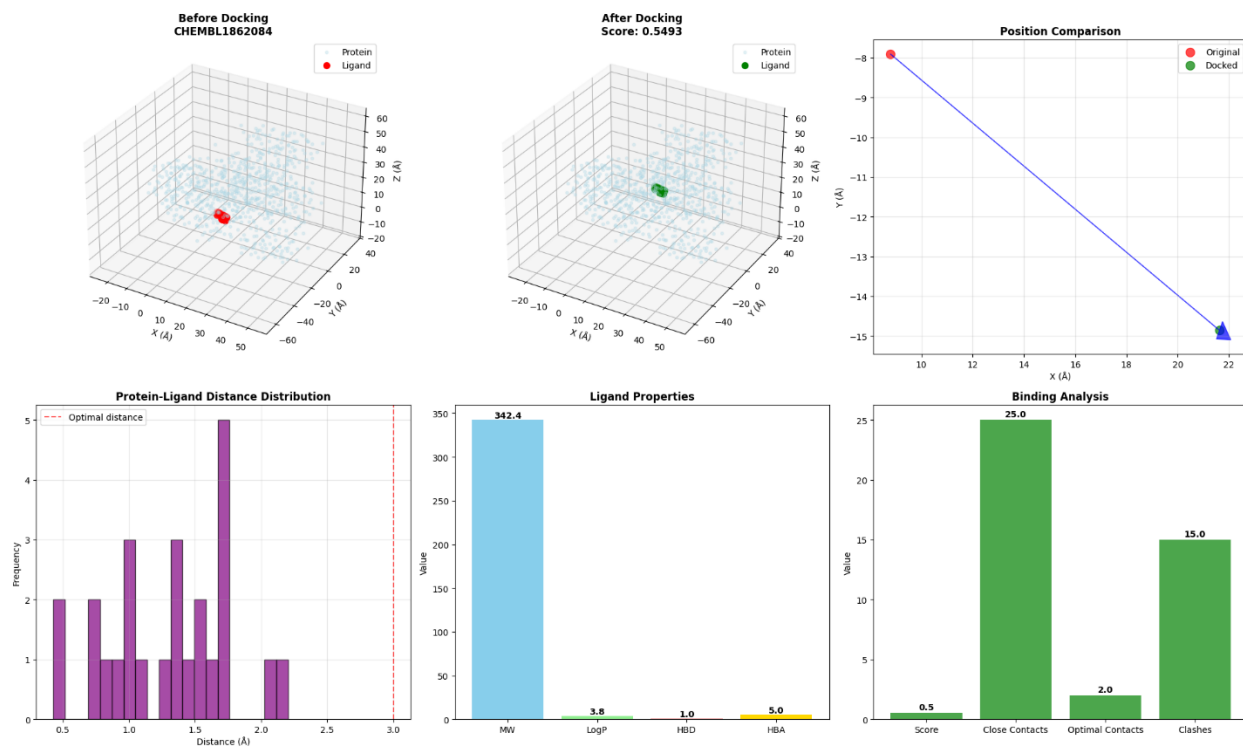
- Docking Score: 0.5887
- Rotate and zoom to explore the binding site

✅ Docking completed - Score: 0.5887

💊 Docking CHEMBL1862084

SMILES: COc1cc(CNCc2coc(-c3cccc(F)c3)n2)cc(OC)c1...

MW: 342.4



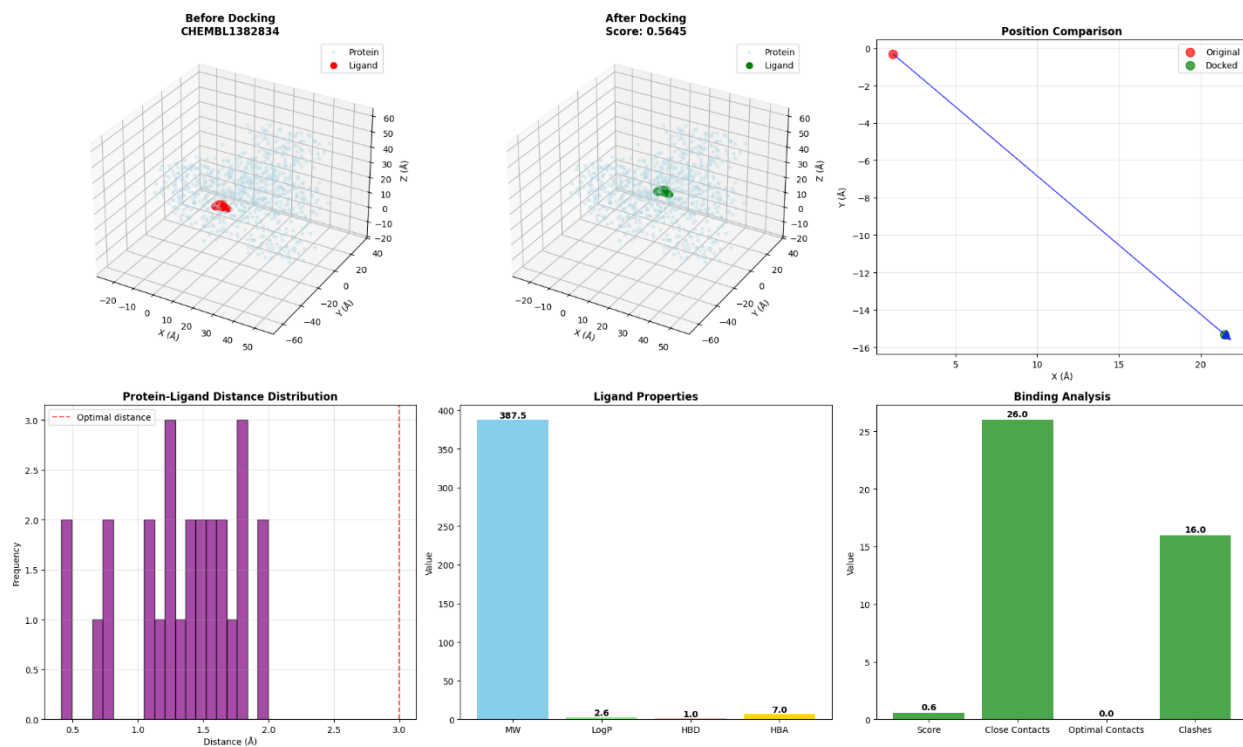
Interactive 3D Visualization for CHEMBL1862084

- Docking Score: 0.5493
- Rotate and zoom to explore the binding site
- ✓ Docking completed - Score: 0.5493

Docking CHEMBL1382834

SMILES: COc1ccc2cc(/C=C3\SC(=S)NC3=O)c(N3CCOCC3)nc2c1...

MW: 387.5



Interactive 3D Visualization for CHEMBL1382834

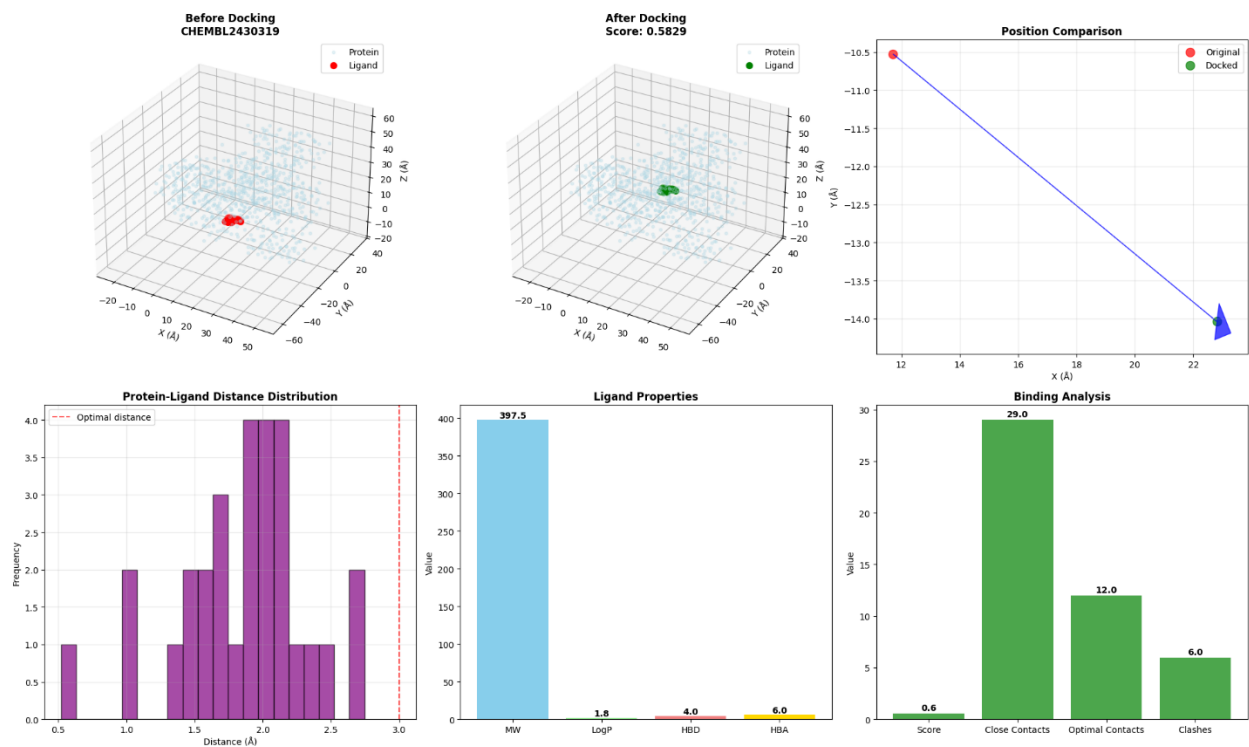
- Docking Score: 0.5645
- Rotate and zoom to explore the binding site

 Docking completed - Score: 0.5645

Docking CHEMBL2430319

SMILES: CCc1ccc(Cc2cn([C@@H]3O[C@H](CO)[C@@H](O)[C@H](O)[C@H]3O)C(=O)N2)cc1

MW: 397.5



Interactive 3D Visualization for CHEMBL2430319

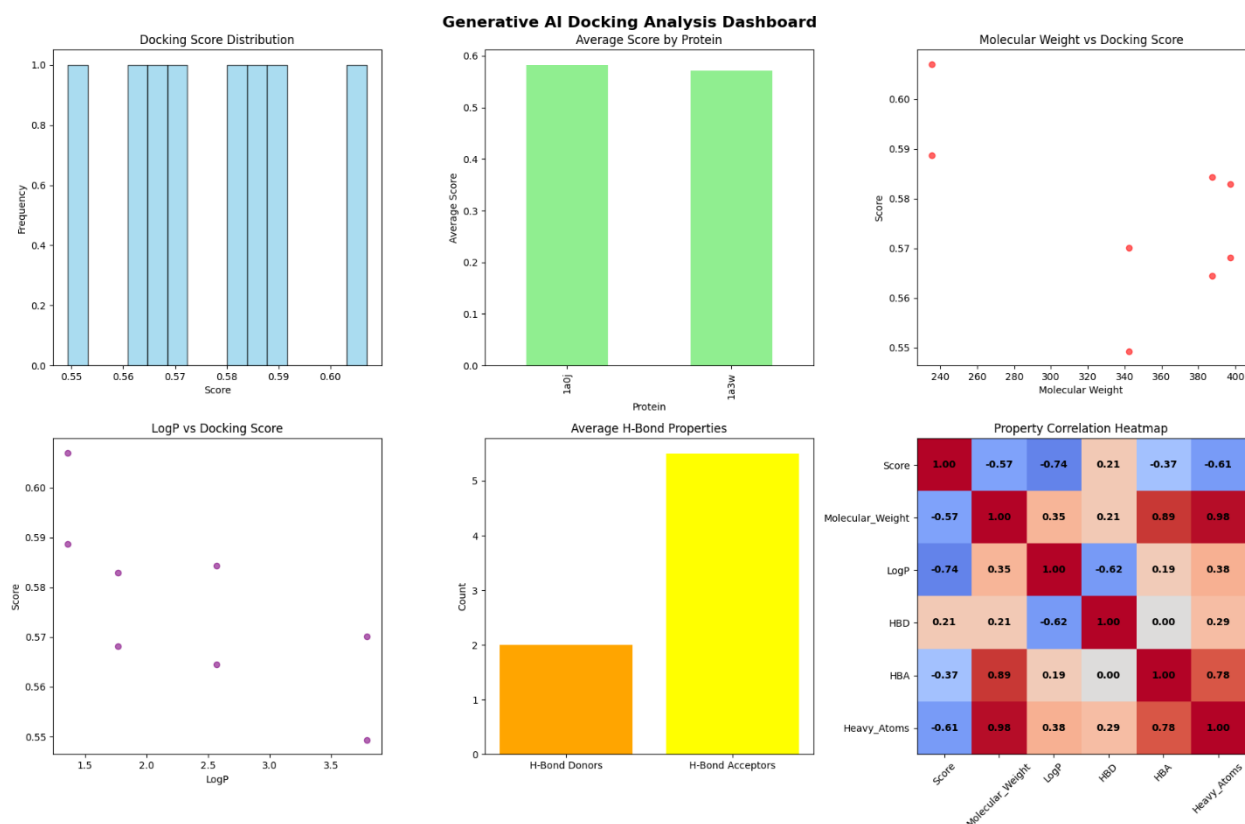
- Docking Score: 0.5829
- Rotate and zoom to explore the binding site
- ✓ Docking completed - Score: 0.5829

COMPREHENSIVE DOCKING ANALYSIS REPORT

🏆 RANKED DOCKING RESULTS:

Protein	Ligand	Score	Molecular_Weight	LogP	HBD	HBA	Heavy_Atoms
1a0j	CHEMBL153534	0.607011	235.316	1.36192	2	4	16
1a3w	CHEMBL153534	0.588721	235.316	1.36192	2	4	16

1a0j	CHEMBL1382834	0.584363	387.486	2.56880	1	7	26
1a3w	CHEMBL2430319	0.582938	397.471	1.76690	4	6	29
1a0j	CHEMBL1862084	0.570082	342.370	3.78770	1	5	25
1a0j	CHEMBL2430319	0.568201	397.471	1.76690	4	6	29
1a3w	CHEMBL1382834	0.564490	387.486	2.56880	1	7	26
1a3w	CHEMBL1862084	0.549327	342.370	3.78770	1	5	25



 Results saved to: generative_ai_docking_results_20250928_224105.csv